Queen Mary
University of London

EBU5376
Digital Signal Processing

Week 4 – Tutorial 1

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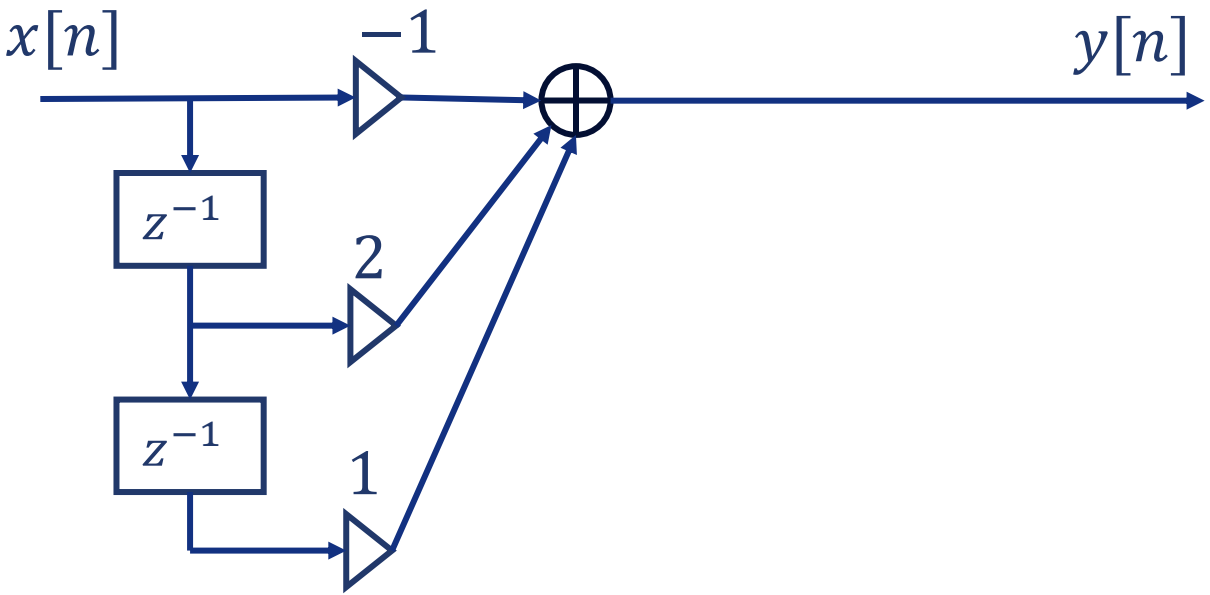
Tutorial Outline

- Block Diagram
- Filter analysis
  - Step Response
  - Amplitude Response
- Evaluation of Amplitude Response
  - Low-pass/High-pass/Band-pass/Band-stop

Tutorial Question 1a

Based on the following block diagram, write down the difference equation of the filter.

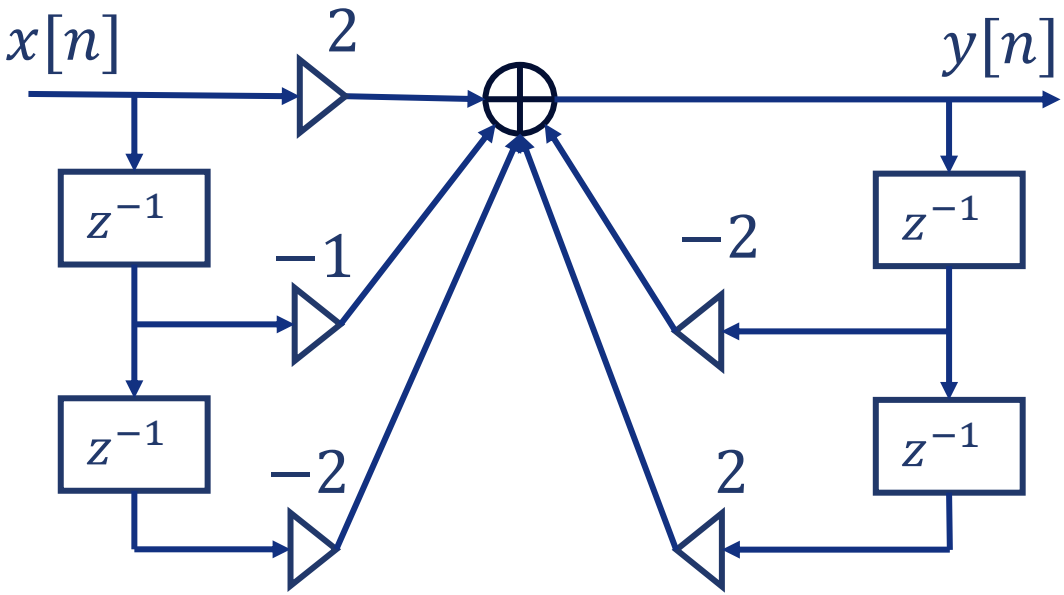


Tutorial Solution 1a

$$y[n] = -x[n] + 2x[n-1] + x[n-2]$$

Tutorial Question 1b

Based on the following block diagram, write down the difference equation of the filter.



Tutorial Solution 1b

$$y[n] = -2y[n-1] + 2y[n-2] + 2x[n] - x[n-1] - 2x[n-2]$$

Tutorial Question 1c

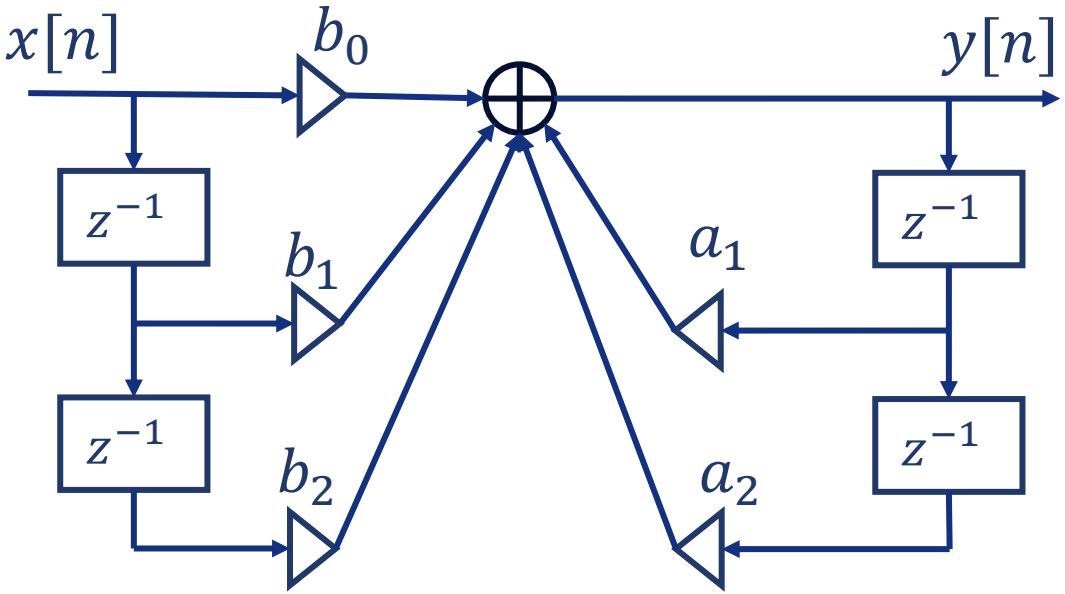
Figure below shows the general block diagram of a filter, where b_0, b_1, b_2, a_1, a_2 are unknown variables. Determine the values of these variables for the following filters:

i. $y[n] = 2x[n] + x[n-1]$

ii. $y[n] = y[n-1]$

iii. $y[n] = y[n-1] + x[n-1]$

iv. Any finite impulse response (FIR) filter.



Tutorial Solution 1c

i. $y[n] = 2x[n] + x[n-1]$

$$a_1 = 0, a_2 = 0, b_0 = 2, b_1 = 1, b_2 = 0$$

ii. $y[n] = y[n-1]$

$$a_1 = 1, a_2 = 0, b_0 = b_1 = b_2 = 0$$

iii. $y[n] = y[n-1] + x[n-1]$

$$a_1 = 1, a_2 = 0, b_0 = 0, b_1 = 1, b_2 = 0$$

iv. Any finite impulse response (FIR) filter.

$$a_1 = 0, a_2 = 0$$

Tutorial Question 1d

Plot the block diagram of the following filters:

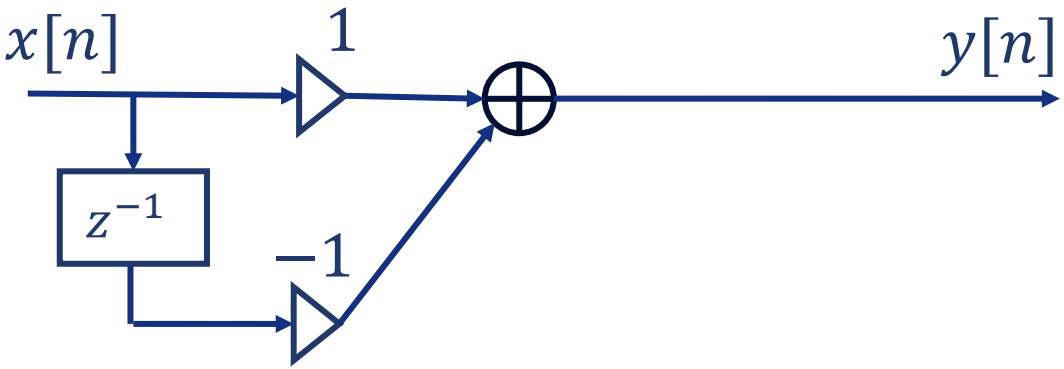
i. $y[n] = x[n] - x[n-1]$

ii. $y[n] = y[n-1]$

iii. $y[n] + 2y[n-1] = x[n-2]$

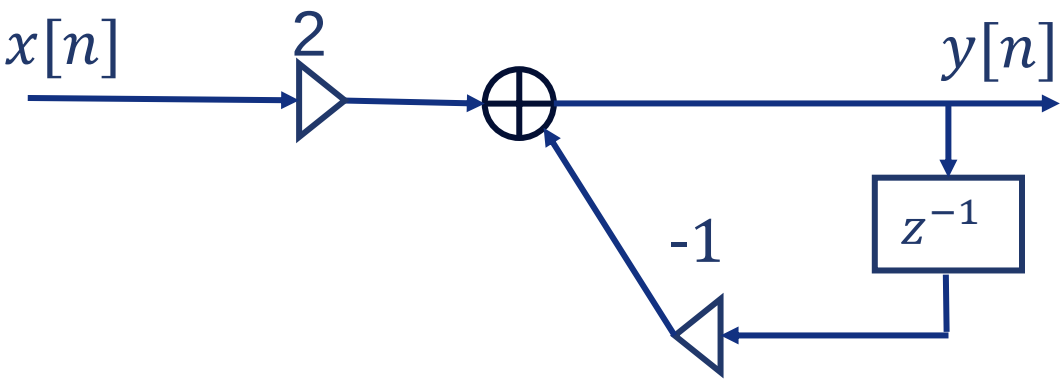
Tutorial Answer 1d

i. $y[n] = x[n] - x[n-1]$



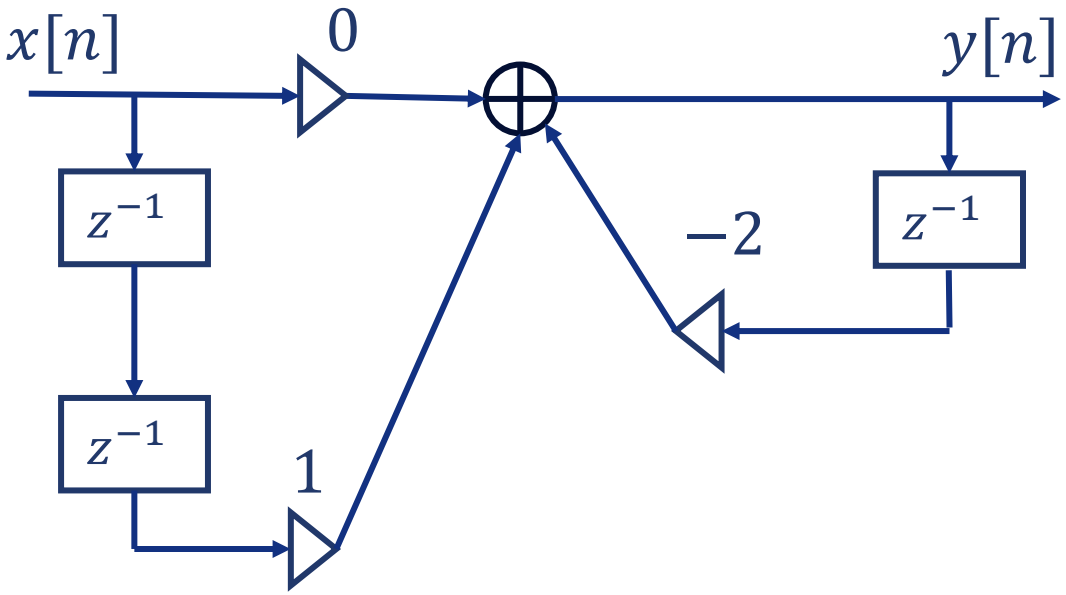
Tutorial Answer 1d

ii. $y[n] = 2x[n] - y[n-1]$



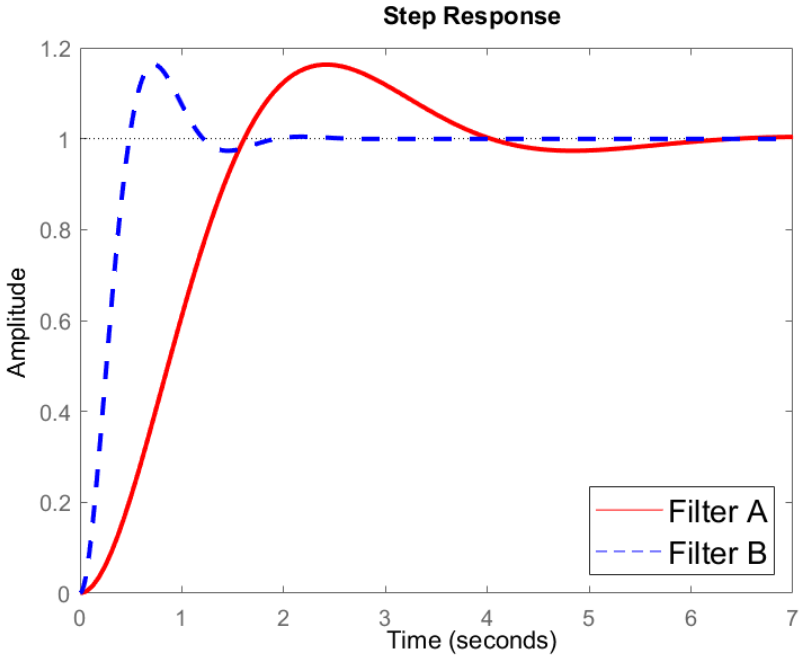
Tutorial Answer 1d

iii. $y[n] + 2y[n-1] = x[n-2]$



Tutorial Question 2a

Figure below shows the step responses of Filter A and Filter B. Based on the figure, describe and compare the time-domain behaviours of the two filters.



Tutorial Answer 2a

The numbers are rough approximations. You will not be asked to calculate the exact values of overshoot and settling time

Filter A:

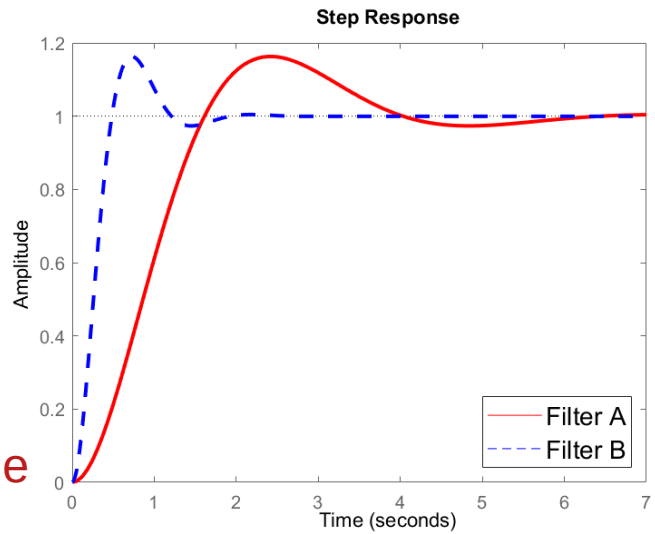
- Overshoots (Peak amplitude ~1.2 > steady state ~1.0)
- Settling time is around ~1.5 sec

Filter B:

- Overshoots (Peak amplitude ~1.2 > steady state ~1.0)
- Settling time is around ~4 sec

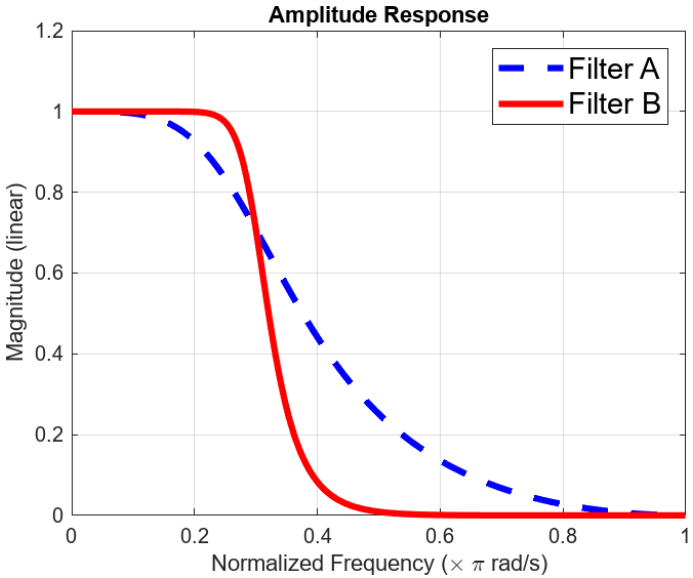
Comparison:

- Filter A has a longer settling time than Filter B, which can lead to delayed responses
- Both exhibit overshoots, leading to distortion of the information



Tutorial Question 2b

Figure below shows the amplitude responses of Filter A and Filter B. Based on the figure, describe and compare the frequency-domain behaviours of the two filters.



Tutorial Answer 2b

Filter A:

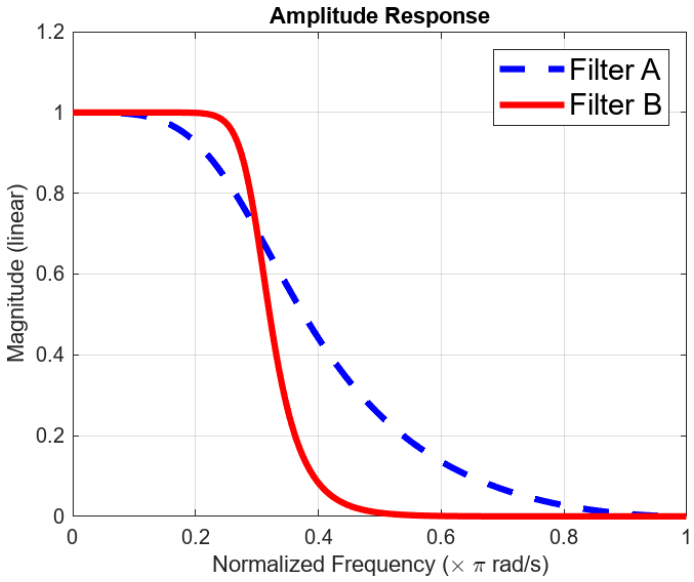
- Amplitude decreases after 0.2π rad/s normalized frequency and approaches zero after 0.9π rad/s normalized frequency,

Filter B:

- Amplitude decreases after 0.3π rad/s normalized frequency and approaches zero after 0.4π rad/s normalized frequency,

Comparison:

- Both are low-pass filters
- Filter A has a longer transition band and a slower roll-off compared to B
- Filter B is better than A at eliminating higher frequencies from the input



Tutorial Question 3a

Consider this filter $y[n] - 0.3y[n-1] = x[n]$

1. Compute the amplitude response

2. Evaluate the amplitude response for frequencies
$$\omega = [0, 0.2\pi, 0.4\pi, 0.6\pi, 0.8\pi, \pi]$$

3. State whether this is a low-pass / band-pass / high-pass / band-stop filter

Tutorial Answer 3a

1. Transfer function:

$$H(z) = \frac{1}{1 - 0.3z^{-1}}$$

Frequency response ($z = e^{j\omega}$):

$$H(\omega) = \frac{1}{1 - 0.3e^{-j\omega}}$$

Amplitude response:

$$|H(\omega)| = \sqrt{H(\omega)H^*(\omega)} = \sqrt{\frac{1}{1 - 0.3e^{-j\omega}} \cdot \frac{1}{1 - 0.3e^{j\omega}}} = \cdots$$

$$= \frac{1}{\sqrt{1.09 - 0.6\cos(\omega)}}$$

Tutorial Answer 3a

2. Substitute $\omega = [0, 0.2\pi, 0.4\pi, 0.6\pi, 0.8\pi, \pi]$ into $|H(\omega)| = \frac{1}{\sqrt{1.09 - 0.6\cos(\omega)}}$, we get $[1.4286 \quad 1.2861 \quad 1.0514 \quad 0.8855 \quad 0.7967 \quad 0.7692]$

3. Since the amplitude response decreases as frequency increases, this is a low-pass filter

Tutorial Question 3b

Consider this filter

$$y[n] + 0.5y[n-1] + 0.2y[n-2] = 0.2x[n] - 0.2x[n-1]$$

Determine whether this is a low-pass / band-pass / high-pass / band-stop filter

Tutorial Answer 3b

Transfer function:

$$H(z) = \frac{0.2 - 0.2z^{-1}}{1 + 0.5z^{-1} + 0.2z^{-2}}$$

Frequency response ($z = e^{j\omega}$):

$$H(\omega) = \frac{0.2 - 0.2e^{-j\omega}}{1 + 0.5e^{-j\omega} + 0.2e^{-2j\omega}}$$

Amplitude response:

$$|H(\omega)| = \sqrt{H(\omega)H^*(\omega)} = \sqrt{\frac{0.2 - 0.2e^{-j\omega}}{1 + 0.5e^{-j\omega} + 0.2e^{-2j\omega}} \cdot \frac{0.2 - 0.2e^{j\omega}}{1 + 0.5e^{j\omega} + 0.2e^{2j\omega}}}$$

$$= \cdots = \sqrt{\frac{0.08 - 0.08\cos(\omega)}{1.29 + 1.2\cos(\omega) + 0.4\cos(2\omega)}}$$

Tutorial Answer 3b

Evaluate using different values of ω in the range $[0, \pi]$

Evaluate $H(\omega)$ for $\omega = [0, 0.2\pi, 0.4\pi, 0.6\pi, 0.8\pi, \pi]$ gives:

$$[0 \quad 0.0800 \quad 0.2033 \quad 0.4193 \quad 0.5717 \quad 0.5714]$$

The amplitude response decreases as frequency increases, hence this is a high-pass filter

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